

Atlas of halving: what one refinement step does, and what emerges, in each dimension

Purpose, 2026-09-27 (user direction: a collective understanding of the halving of the lattice and what emerges in each case and dimension, as the mid-term goal). This note is the shared map. Each cell says what one halving does, what it leaves behind, and what survives the limit, with the status of the statement and the note that holds it. It adds no new theorem; it fixes one vocabulary so that the agents working here (Claude, GPT-6 Astra, Fable reviewers) and the user fill the same cells. Open cells are listed in §5.

Status labels: **proved** (theorem in a repository note, reviewed or not as marked), **exact** (closed-form identity), **formal** (derivation assuming an unproved estimate), **known** (established literature, cited), **open**.

1. The operation

Halving one direction $a_1 \mapsto a_1/2$ of a lattice with plaquette heat times $t_{\mu\nu} = \lambda_D a_\mu a_\nu / \prod_{\rho \neq \mu, \nu} a_\rho$ (series/parallel note, eq. (1), small-field regime) splits exactly into two moves.

- **Series move.** A face containing the refined direction is cut in two; its heat time halves. Integrating the cutting edge is a convolution of the two half-weights: character coefficients multiply. For heat-kernel weights it closes exactly in every dimension.
- **Parallel move.** A face transverse to the refined direction gets a new copy in the mid-plane, both at doubled heat time. Integrating the mid-plane leaves the factor Ψ : a $(D-1)$ -dimensional gauge theory whose edges carry Brownian-bridge laws fixed by the coarse field. Pointwise products of weights add their logarithms; the heat kernel is closed under this only up to vortex terms (zero-spacing note, Proposition 1).
- **Count.** A halving produces parallel moves in $\binom{D-1}{2}$ transverse planes. An isotropic step is D directional halvings; the isotropic heat time scales as $t \propto \lambda_D a^{4-D}$.
- **Time-only halving** (Euclidean time at fixed spatial lattice) sends the transverse heat times to infinity; the Hamiltonian limit needs exponential (Wilson-type) magnetic weights (zero-spacing note, §1).

Newton's halving is the one-dimensional case: inserting a time $t_{2,6}$ in a cell is a series move (refinement note, §2). A physical record at the inserted time is a new variable attached to the body, the analogue of a parallel insertion. Forgetting a Gaussian record is an exact parallel move with defect proportional to $\hbar^2$, the identity for $\hbar = 0$ (record as parallel move).

1b. Cuts at any position (user remark, 2026-09-27)

Halving is the easiest cut; any cut at fraction $s \in (0, 1)$ of a cell performs the same two moves, with s as a parameter.

- **Series move.** The cut face splits into heat times st and $(1-s)t$; the convolution returns t exactly, for heat-kernel weights, in every dimension.
- **Parallel move.** In $D = 3$ the transverse copies get heat times t/s and $t/(1-s)$; at the quadratic level precisions add, $s/t + (1-s)/t = 1/t$. The inserted edges carry the bridge law observed at fraction s , variance $s(1-s)t_e$ in place of $t_e/4$. Halving maximizes the bridge variance, so each non-symmetric cut is gentler than a halving; the s -dependence of the defect bounds (Prop. 2, Theorem 5) is expected through $4s(1-s)$ and has not been derived.

- **Newton.** The cell action $K_\tau = F^2\tau^3/(24m)$ (two-path note) obeys $\tau^3 = (s\tau)^3 + ((1-s)\tau)^3 + 3s(1-s)\tau^3$, so one cut removes exactly $3s(1-s)K_\tau$, at most $\frac{3}{4}K_\tau$ (halving). Along any sequence of cuts whose mesh tends to zero the removed amounts sum to K_τ , whatever the positions and order (the remainder is $\sum_i C\tau_i^3 \leq C\tau \max_i \tau_i^2$). The Galileo action is therefore an additive measure on the cut process: the refinement limit is cut-independent, and every cut spends a definite share of the action that Newton sends to zero. This is the Lévy–Ciesielski structure of the Brownian bridge (variance $s(1-s)$ at each dyadic or non-dyadic insertion) seen in the action.
- **The frame.** A cut at an arbitrary, even irrational, position is the geometers’ cut of Book I’s closing scholium. The questions for each cell become: does the limit depend on the cut sequence (for the free field and for Newton, no), and does the universal part (the $\log 2$ of cell 4, which for general cuts should read $\log(1/s)$ summed over the step) survive when s varies (conjecture: yes, as the scheme-independent one-loop coefficient).

2. The atlas by dimension

		Parallel planes per halving	What one step leaves	What survives the limit	Status and notes
$1+n$	Series move				
$0+0$	none (a single weight)	none	the weight $k_t(U)$ itself; a single integral with an $e^{-c/\hbar}$ structure	a large- N transition (Gross– Witten)	known; dimension ladder
$1+0$ Newton	exact after one cubic counterterm $-F^2\hbar^3/(24M)$; the defect $-F^2uv\hbar/(8M)$ is a coboundary	none	a scalar	unobserved: the exact propagator at every $\hbar$; recorded: a floor only if joint determinacy fails	proved; refinement note, fifth postulate
$1+1$	exact (heat-kernel convolution)	0	nothing for heat kernels; for any action, $a^{-2}(1-\hat{c}_R)$	a conjugation- invariant Lévy exponent $\psi(R)$; Yang–Mills iff Lindeberg; circle spectrum $\hbar cL\psi(R)$; no local particle	proved (Theorem 2 of the zero-spacing note); known (Lévy)

$1 + n$	Series move	Parallel planes per halving	What one step leaves	What survives the limit	Status and notes
$1 + 2$	exact	1	free: an irrelevant quadratic form of relative size $a^2 K^2/16 +$ $(Ga)^2/8$; $U(1)$: vortices of density $e^{-\pi^2/(8t)}$; $SU(N)$: coupling shifts $O(t)$, curvature term explicit	free photon for $U(1)$ at fixed λ_3 (every lattice gap dies); conjectured gap $C_3 \hbar c \lambda_3$ for $SU(N)$	proved for free and $U(1)$; formal for $SU(N)$ (Proposition 7); see §3
$1 + 3$	exact	3	$t = g^2$ fixed: coupling shifts that recur each step; large fields suppressed only as $(a\Lambda)^{2b_0c}$	running coupling, $\Lambda =$ $\mu e^{-1/(2b_0g^2)}$; small instantons die ($11N/3 > 4$); dislocations need $c > 2/b_0$ ($\frac{6}{11}$ of an instanton for $SU(2)$); $U(1)$: Coulomb phase	known (running, 6/11 criterion, Driver); the measure-level route fails for $U(1)$ (proved)
$1 + n, n \geq 4$	exact	$\binom{n}{2}$	t grows as $a \rightarrow 0$	no small-field regime; a phase transition, no established continuum limit	known (Creutz for $SU(2)$ in 5D)

The same heat time $t = \hbar g_{\text{cl}}^2 a^{4-D}$ controls the column “what one step leaves”. For $D < 4$ refining and taking $\hbar \rightarrow 0$ push it the same way; in mechanics refining makes each cell more quantum (dimension ladder, §2).

3. The atlas for $1 + 2$ by group

Group	Isolated cube	Full mid-plane	One step, measure level	Iteration and limit
free ($\mathbb{R}^n$)	exact	exact defect (Proposition 2 of the series/parallel note)	Gaussian, exact	blocked actions converge to a Gaussian fixed point (known, Bell–Wilson; for this blocking asserted)
$U(1)$	exact (Proposition 3)	exact charge form, vortex bound (Proposition 4, Corollary)	one step equals the free step up to density $e^{-\pi^2/(8t)}$ (Theorem 5, refereed); whole measure within TV $2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}$ of the monopole-free part (monopole note)	free photon (Gross 1983, known)
$SU(2)$	midpoint characters exact for every spin; spin- $\frac{1}{2}$ and spin-1 cube sectors exact, with matrix image bounds for spin 1 (closed form, Theorems 1–4)	formal order- t expansion: cut shifts positive, transverse negative; other local operators have dimension at least six; explicit torus winding term	formal; needs normalized estimate (18) with winding control	open
$SU(3)$	curvature term explicit, softening $t_j X_j ^2/512$ (group-general form of Proposition 6)	open	formal	open; the gap $C_3 \hbar c \lambda_3$ is the infrared obligation

4. What emerges, read across the atlas

- **Series moves never produce anything new.** In every dimension they close exactly for heat-kernel weights, and in mechanics after one scalar counterterm. The one-dimensional Newton problem and the two-dimensional gauge theory are all series.
- **Parallel moves carry everything that renormalizes.** Their number, $\binom{D-1}{2}$, is zero exactly where the theory is exactly soluble.
- **The exponential ladder.** Each compact correction left by a step is $e^{-c/t}$ (vortices, monopoles, dislocations). In $D < 4$ it is summable per physical volume at fixed coupling, so compact effects die; in $D = 4$ the running turns it into a power of a , so survival is a

threshold condition; in $D > 4$ there is no small t .

- **What survives is trajectory-dependent.** Compact $U(1)$ in $1 + 2$ keeps a gap along $\lambda_3 a \simeq c_0/(2 \log(1/a))$, $c_0 \approx 4.99$ the monopole exponent, where the monopole density per physical volume diverges, and loses it at fixed λ_3 (Göpfert–Mack versus Gross).
- **Constants that emerge.** Mechanics: one scalar counterterm per cell, and one action constant if joint determinacy is denied (Theorems B and B' of the fifth-postulate note). $1 + 1$: the Lévy exponent (one coupling for Yang–Mills). $1 + 2$: one coupling, shifted summably. $1 + 3$: one scale Λ by dimensional transmutation.

5. Open cells

1. $SU(2)$ cube: *filled through* $J = 1$ (Theorem 4, exact diagonal entries, image bounds and cube sector); $J > 1$ entries and control of the full spin sum remain open.
2. $SU(2)$ full mid-plane in $1+2$: formal order- t calculation filled, refereed (all items accepted): cut shifts positive, transverse negative, remaining bulk terms of dimension six; the fixed-torus winding term is removed by the clause $\min_j N_j \geq c_0/t$; open: the uniform estimate (18).
3. Stability of Hypothesis $P(\alpha)$ under iteration. *For* $U(1)$ *in* $1 + 2$, *reduced to exact Gaussian blocking* (Corollary 2' of the monopole note); open for $SU(N)$.
4. $1 + 3$: the one-loop coefficient $-2b_0 \log 2$ per isotropic step from the three parallel insertions of each halving.
5. Time-only halving. *Structural reading, 2026-09-27, of known results.* At fixed spatial lattice the electric faces make the series moves: heat-kernel electric weights are a semigroup in the time step, so they compose exactly. The magnetic faces make the parallel moves: with exponential weights $e^{-a_0 V}$ they close exactly in their own family, since products of such weights add the exponents. The whole defect of a time halving is the non-commutation of the two families, and it vanishes as $a_0 \rightarrow 0$ by the Trotter product formula (Trotter 1959; Chernoff 1968; metadata; strong convergence for the Laplacian on G^E plus a bounded magnetic potential). This is why the Hamiltonian limit (Kogut and Susskind 1975; transfer matrix: Creutz 1977; metadata) exists at every fixed spatial lattice, while spatial halving leaves the non-closing factor Ψ . Open: an operator-norm rate, which needs domain estimates for the commutator of the Laplacian with V .
6. *Filled 2026-09-27 for Gaussian records* (record as parallel move): forgetting a record convolves momentum with variance $\hbar^2/(4\sigma^2)$; precisions add in parallel; refinement converges iff $\sum \sigma_j^{-2} < \infty$; identity for $\hbar = 0$.
7. The infrared: from ultraviolet control in $1 + 2$ to $C_3 > 0$.

6. Consequence for STATE

STATE carries the mid-term goal and points here. Each result that fills a cell updates the cell in this note and nothing else; the notes that prove results remain the record.